

Date & Time

Alston Yam

Learning Goals

1 Date

Here are a few useful things to know:

1. A normal year has 365 days, a leap year has 366 (February would have 29 days).
2. Every 4 years is a leap year, every 100 years is not a leap year, every 400 is a leap year.
3. January 1st 2001 was a Monday. This means that January 1st 2002 will be a Tuesday. (This is different for leap years).

Lets explain point 2 a bit more. Normally, if a year number is divisible by 4, it is a leap year (do you remember what is the divisibility rule for 4)? But if that year is a multiple of 100, it **isn't** a leap year, but if the year number is a multiple of 400, it is a leap year.

Problem 1.1 Is 1736 a leap year?

Problem 1.2 Is 2014 a leap year?

Problem 1.3 Is 1800 a leap year?

Problem 1.4 Is 2000 a leap year?

Problem 1.5 Is 1200 a leap year?

Problem 1.6 Is 542 a leap year?

Problem 1.7 Today is Sunday 4th January, 2026. What day of the week will it be exactly two years from now?

Problem 1.8 If today is a Thursday, what day of the week would it be 50 days from now?

Problem 1.9 For a certain leap year, January 21st is a Wednesday. What day of the week would January 21st the next year be?

Problem 1.10 Alex was born on Monday January 1st, 2001. Suppose today is his 15th birthday. What day of the week is today?

2 Time

Standard stuff:

1. 1000 milliseconds are in a seconds
2. 60 seconds in a minute
3. 60 minutes in an hour
4. 24 hours in a day

Problem 2.1 How many seconds are in a day?

Problem 2.2 How many hours are in a week

Problem 2.3 – Quite Hard Suppose I have been alive for 1 billion seconds. How old am I, in years?

3 Problems

Problem 3.1 This year Galileo would've been 462 years old. How old was he if he died in 1642? (This year is 2026)

Problem 3.2 The three Smith children have all moved out of home. Sasha returns every 10 days to have dinner with her parents. Connor returns every 4 days for dinner and Kayla returns every 12 days to eat with her parents. If all three children have dinner with their parents on Sunday, what day of the week will it be the next time the three children are all back home for dinner?

Problem 3.3 In a certain year, there were exactly four Tuesdays and exactly four Fridays in October. On what day of the week did Halloween, October 31st, fall on that year?